

NATALIE R. MYERS, PhD

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Education:

2025	Ph.D. Physics	Astrophysics Track, Texas Christian University (TCU), Fort Worth, TX “ <i>Tracing the Chemical Enrichment of the Milky Way Using Star Clusters: Measurement and Analysis of Neutron Capture Elemental Abundances</i> ”
2022	M.S. Physics	Texas Christian University (TCU), Fort Worth, TX “ <i>Measuring Galactic Chemical Abundance Trends in Open and Globular Clusters Using SDSS/APOGEE</i> ”
2020	B.S. Physics	University of St. Thomas (UST), St. Paul, MN (Minor: Math)

Employment and Affiliation:

2025–Present	Assistant Research Scientist,	Johns Hopkins University, <i>Supervisor:</i> K. Schlaufman
2021–2025	Research Assistant,	Texas Christian University, <i>Advisor:</i> P. M. Frinchaboy
2020–2025	Teaching Assistant,	Texas Christian University
2019–2020	Physics Tutor,	University of St. Thomas
2018–2020	Observatory Crew Member,	University of St. Thomas

Teaching Experience

TA and Tutoring Experience:

2025 Spring	PHYS 10164:	Classical Physics 2 Lab	(2 sections)
2024 Spring	PHYS 10164:	Classical Physics 2 Lab	(3 sections)
2021 Summer	PHYS 10164:	Classical Physics 2 Lab	(1 section)
2021 Spring	PHYS 10263:	Cosmic Origins Lab	(2 sections)
2020 Fall	PHYS 10273:	Intro to Astronomy: Earth & Planets Lab	(2 sections)
2019-2020	Tutoring:	Up to Calculus III and Modern Physics	

Workshops and Guest Lectures:

2026 Summer		John’s Hopkins Teaching Academy: Teaching Institute <i>Pedagogy-focused; inc. active learning, course design, effective assessment, etc</i>
2026 Spring	AS.171.611:	Stellar Structure and Evolution lecture <i>Covered graduate level lecture on nucleosynthesis</i>
2023 Fall	PHYS 10273:	Cosmic Origins guest lecture: Potential for Life in the Solar System
2023 Fall	PHYS 30123:	Observational Physics Lab Assistant (2 nights) <i>Operated 2.7-m Telescope during class trip to McDonald Obs.</i>

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Leadership Experience

Co-Mentoring Graduate Students

2025–2026	R. Bachchan (TCU)	Taught spectroscopic methods and how to use BACCHUS
2024–2026	A. Sinha (Univ. of Utah)	Assisted in BACCHUS analysis training
2024	J. Otto (TCU)	Trained on the 2.7-m telescope at McDonald Observatory
2023	A. Wiggins (TCU)	Trained on the 2.7-m telescope at McDonald Observatory

Mentoring Undergraduate Students

2026	Mentored Johns Hopkins Rowland Summer Research Fellow: Jalane Mendo	
2023	Supervised REU student: H. Wallace	
2023	Co-mentored REU student: K. Toguchi-Tani	
2022	Co-Mentored REU student: K. Thomas	
2018–2019	Observatory Worker	
	– Trained introductory astronomy students to operate UST 0.4-m telescope (4 semesters)	
	– Gave tours for star parties and other community outreach events	

Grants, Awards, and Fellowships:

2026	AAS International Travel Grant (\$2000??)
2024	AAS International Travel Grant (\$998)
2024	Sigma Xi Grant in Aid of Research (\$2,765)
2024	Graduate Travel Grant, TCU (\$400, \$800)
2023	AAS 241 Chambliss Astronomy Achievement Award Winner
2020–2021	Ida Green Fellowship, TCU (\$56,010)
2019–2020	MN Space Grant Consortium Award for Significant Contributions to Research (\$500)
2019	Young Scholars Research Grant, Advisor: M. Wood

Observing Experience (Awarded):

2026Q2	APO 3.5-m/ARCES	(PI; 2 half nights)
2026Q1	APO 3.5-m/ARCES	(PI; 2 half nights)
2024B	McDonald 2.7-m/Coudé	(Co-I; 3 nights)
2024A	Keck 10-m/HIRES	(Co-I; 9 half nights)
2023C	McDonald 2.7-m/Coudé	(PI ; 6 nights)
2023C	McDonald 2.7-m/Coudé	(PI ; 6 nights)
2023B	Magellan 6.5-m/MIKE	(Co-I; 2 nights)
2023B	Lick 2.4-m/APF	(Co-I; 7 nights , queue scheduled)
2023B	Lick 3.0-m/Hamilton echelle	(Co-I; 3 nights)
2023B	Keck 10-m/HIRES	(Co-I; 1 night)
2023B	McDonald 2.7-m/Coudé	(Co-I; 4 nights)
2023A	Magellan 6.5-m/MIKE	(Co-I; 2 nights)
2022B	McDonald 2.7-m/Coudé	(Co-I; 4 nights)
2022A	Magellan 6.5-m/MIKE	(Co-I; 2 nights)

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Other Professional Experience and Volunteering

- Volunteer at the JHU Annual Physics Fair (Spring 2026)
 - TAC committee member (2026)
 - Astronomy & Astrophysics Journal Referee (2025,2026)
 - Graduate Student Representative at TCU (2023-2024)
 - CUWiP Conference Local Organizing Committee (LOC) member and volunteer, (Jan 2023 at TCU)
 - Volunteering for Atatiana Projects 2023 STEAM summer camp (July 2023)
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Other Skills

Python • Data Science • MATLAB • Microsoft Office • OOP • Science Communication
GitHub • Spectral Analysis • LaTeX • Observational Astronomy • SQL

NATALIE R. MYERS – PUBLICATIONS

Refereed Publications

- SDSS Collaboration, Adamane Pallathadka, G., Aghakhanloo, M., Aird, J., Almeida, A., et al. (212 authors listed alphabetically; including **Myers, N.**), (2025), “*The Nineteenth Data Release of the Sloan Digital Sky Survey*”, *Astronomical Journal*, *submitted*, doi: 10.48550/arXiv.2507.07093.
- Sinha, A., Zasowski, G., **Myers, N. R.**, Manea, C., Frinchaboy, P., et al., (2026), “*The Chemical Homogeneity of Single-Lined Spectroscopic Binaries in Open Clusters*”, *MNRAS*, 547, 440, doi: 10.1093/mnras/stag440.
- Otto, J. M., Frinchaboy, P. M., **Myers, N. R.**, Johnson, J. W., Donor, J., et al., (2025), “*The Open Cluster Chemical Abundances and Mapping Survey: VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*”, *Astronomical Journal*, 171, 91, doi: 10.3847/1538-3881/ae28d8.
- Kollmeier, J. A., Rix, H.-W., Aerts, C., Aird, J., Alfaro, P. V., et al. (220 authors total, including **Myers, N.**), (2025), “*Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy*”, *Astronomical Journal*, 171, 52, doi: 10.3847/1538-3881/ae0576.
- Spoo, T., Thomas, K., “Kaleo” Toguchi-Tani, E., Otto, J., **Myers, N.**, et al., (2025), “*Extending the [C/N]-Age Calibration: Using Globular Clusters to Explore Older and Metal-Poor Populations*”, *Astronomical Journal*, 170, 333, doi: 10.3847/1538-3881/ae14ec.
- Schiavon, R. P., Phillips, S. G., **Myers, N.**, Horta, D., Minniti, D., et al., (2024), “*The APOGEE value-added catalogue of Galactic globular cluster stars*”, *MNRAS*, 528, 1393, doi: 10.1093/mnras/stad3020.
- Almeida, A., Anderson, S. F., Argudo-Fernández, M., Badenes, C., Barger, K., **et al.** (155 authors listed alphabetically; including **Myers, N.**), (2023), “*The Eighteenth Data Release of the Sloan Digital Sky Surveys: Targeting and First Spectra from SDSS-V*”, *ApJS*, 267, 44, doi: 10.3847/1538-4365/acda98.
- Myers, N.**, Donor, J., Spoo, T., Frinchaboy, P. M., Cunha, K., et al., (2022), “*The Open Cluster Chemical Abundances and Mapping Survey. VI. Galactic Chemical Gradient Analysis from APOGEE DR17*”, *Astronomical Journal*, 164, 85, doi: 10.3847/1538-3881/ac7ce5.
- Spoo, T., Tayar, J., Frinchaboy, P. M., Cunha, K., **Myers, N.**, et al., (2022), “*The Open Cluster Chemical Abundances and Mapping Survey. VII. APOGEE DR17 [C/N]-Age Calibration*”, *Astronomical Journal*, 163, 229, doi: 10.3847/1538-3881/ac5d53.
- Abdurro’uf, Accetta, K., Aerts, C., Silva Aguirre, V., Ahumada, R., **et al.** (341 authors listed alphabetically; including **Myers, N.**), (2022), “*The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data*”, *ApJS*, 259, 35, doi: 10.3847/1538-4365/ac4414

Talks

No Universal Chemical Clock: Testing Age–Abundance Relationships with Subgiant Stars, (Jun. 2026), Cool Stars 23 splinter talk, Tokyo, Japan.

Understanding the Chemical Composition of the Milky Way: A Study of Open Clusters and the Heavy Elements Within, (Nov. 2025), Invited Speaker, CAS Emerging Scholars Seminar, University of St. Thomas, St. Paul, MN.

Neutron Capture Abundances of SDSS-IV/APOGEE-Selected Open Cluster Stars: Galactic Gradients and Applications to Chemical Clocks, (Sep.2025), Wine and Cheese Seminar, Johns Hopkins University, Baltimore, MD.

Neutron Capture Abundances of SDSS-IV/APOGEE-Selected Open Cluster Stars: Galactic Gradients and Applications to Chemical Clocks, (Jan. 2025), AAS245 dissertation talk, 129.06D, National Harbor, MD.

Conference Proceedings

Myers, N., Loebman, S., Reggiani, H., & Frinchaboy, P., (2025), “*Beyond OCCAM: Measuring Optical Neutron Capture Abundances of Open Cluster Stars*”, Proceedings of the IAU Symposium 395, doi: 10.48550/arXiv.2510.13014.

Select Poster Presentations

Myers, N., Frinchaboy, P., Loebman, S., Reggiani, H., Holmbeck, E., Donor, J., Shetrone, M., (2024), “*Neutron Capture Follow-up of SDSS-IV/APOGEE Open Clusters: Galactic Gradients and Applications to Chemical Clocks*”, IAU Symposium 395: “Stellar populations in the Milky Way and beyond”, Paraty Brazil, November 2024

Myers, N., Reggiani, H., Loebman, S., Holmbeck, E., Shetrone, M., Frinchaboy, P., & The Occam Team, (2024), “*Surveying for Ancient Supernovae Across the Galaxy: Mapping SDSS-IV/OCCAM Abundances and the Search for Heavier Elements*”, AAS, 243, 458.07, January 2024.

Myers, N., Frinchaboy, P., Donor, J., Spoo, T., Otto, J., Wiggins, et al., (2023), “*Mapping Chemical Abundances Across the Milky Way with Open Clusters in SDSS/APOGEE: Gradients of Iron and Light Elements plus Neutron Capture Follow-Up*”, Surveying the Milky Way: The Universe in Our Own Backyard, Pasadena CA, October 2023.

Myers, N., Frinchaboy, P., Spoo, T., Wiggins, A., Otto, J., et al., (2023), “*The Open Cluster Chemical Abundances and Mapping Survey Follow-up: Tracing the Neutron-Capture Enrichment of the Milky Way*”, SDSS-V Meeting NYC, August 2023.

Myers, N., ”Painting a Portrait of a Young Milky Way using Globular Clusters”, (2023), Student Research Symposium at TCU, April 2023.

Myers, N., Frinchaboy, P., Donor, J., & Spoo, T., (2023), “*Exploring the Milky Way’s Childhood: Globular Clusters Abundances from SDSS/APOGEE*”, AAS, 241, 402.14.

Myers, N., Frinchaboy, P., Donor, J., Cunha, K., Spoo, T., et al., (2022), “*OCCAM VI: Assembling the open cluster “Avengers” of Galactic Evolution*”, AAS, 240, 201.04, January 2022.

Myers, N., “Assembling the Open Cluster ‘Avengers’ of Galactic Evolution”, Student Research Symposium at TCU, April 2022.
(*Canceled due to COVID-19*).

Myers, N., & Wood, M., (2021), “*Calibrating a New Spectrometer Using Arcturus*”, AAS, 237, 547.18, January 2021 (*Virtual to COVID-19*).

Collaborative Research Presentations

Spoo, T., Frinchaboy, P., Tayar, J., Souto, D., & The Occam Team, (2024), “*Deriving Ages for Field Stars: Tools and Complications from Stellar Evolution*”, AAS, 243, 430.04D.

Otto, J., Spoo, T., Toguchi-Tani, E., Thomas, K., Frinchaboy, P., & The Occam Team, (2024), “*Digging through the Galactic Graveyard: Chemistry and Ages of “Dead” Milky Way Satellite Galaxies*”, AAS, 243, 107.05.

Wiggins, A., Bhattarai, B., Loebman, S., **Myers, N.,** Frinchaboy, P., Otto, J., Spoo, T., & Donor, J., (2024), “*Only the Special Survive: Star Cluster Disruption in Milky Way-like Galaxy Simulations*”, AAS, 243, 458.20.

Wallace, H., **Myers, N.,** Otto, J., Donor, J., Spoo, T., Wiggins, A., & Frinchaboy, P., (2024), “*The Open Cluster Chemical Abundances and Mapping (OCCAM) Survey: First Results from SDSS-V*”, AAS, 243, 458.02.

Wiggins, A. I., Bhattarai, B., **Myers, N.,** Loebman, S., Frinchaboy, P., et al., (2023), “*Only the Special Survive: Star Cluster Disruption in Galaxy Simulations*”, Surveying the Milky Way: The Universe in Our Own Backyard, Pasadena CA.

Otto, J., Wallace, H., Frinchaboy, P., **Myers, N.,** Spoo, T., et al., (2023), “*The Open Cluster Chemical Abundance and Mapping (OCCAM) survey: Cross-calibration of APOGEE to MWM/ASTRA*”, New York City NY.

Other Conferences Attended

- 2023 Conference for Undergraduate Women in Physics at Texas Christian Univ., Fort Worth, TX.
 - 2020 Conference for Undergraduate Women in Physics at Univ. of Minnesota, St. Paul, MN.
 - 2019 Conference for Undergraduate Women in Physics at Northwestern Univ., Chicago, IL.
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